

Individual Reflection

Looking back at this course and both XR projects, I can really feel that I've grown technically. The AR and VR parts challenged me in totally different ways, and I ended up with a much stronger understanding of what XR development actually involves, not in theory, but when you're in Unity trying to make things work in the real world.

AR Project

During this AR project, I mainly worked with AR Foundation and image tracking. It was interesting to see how accurate image tracking can be, yet at the same time so fragile. I suddenly grasped why tracking would depend so much on things like contrast, lighting, and reference image quality. Those are details I knew in theory but physically working with them made things click.

I also spent time on UI/UX work and polishing the user experience. AR easily becomes overwhelming, because you're mixing the real world with virtual objects, so keeping everything simple is actually important. Our idea made sense to me, as it's taking something kids already do - reading or listening to stories - and combining that with something fun and interactive. This didn't feel like "technology for the sake of technology," but rather something that could actually make the experience more engaging.

VR Project

The VR project was a whole other world. I primarily dealt with the weapon system, and I honestly did not expect it to be such a challenge making something simple, like a magazine or a grab interaction, feel natural in VR. It doesn't take much before the gun starts rotating weirdly, clipping through the hand, or causing physics chaos.

I spent a lot of time improving the grabbing so that it'd feel like the gun was stable in the player's hand. That really taught me how much small details matter with regards to immersion. Likewise, with audio, directional sound and all the zombie noises and gunshots add so much to the experience. When a zombie comes from behind and you can hear it, it changes everything.

The biggest challenge was certainly the weapon system and all physics concerning it. Getting the recoil, grabbing, colliders, and shooting to work smoothly took a lot of systematic debugging. Change one thing at a time, test, repeat. It's been tough but also very rewarding.

Understanding XR Theory

One of the theoretical areas that I think I understood much better through this course was tracking both AR image tracking and VR's 6DOF head/hand tracking suddenly made

more sense after having worked with them. You really realize how important accurate tracking is to immersion and why everything breaks when performance drops.

Speaking of which, the workflow taught me a lot, too. VR development isn't just build and run. You have to think about performance constantly. Mobile VR is limited but surprisingly powerful if you optimize properly.

Immersion and User Experience

For me, hands and interaction are what make VR truly immersive. Being able to grab things, load a gun, shoot, throw objects — that's what brings the virtual world to life. That's also why the weapon system felt so important. It wasn't just another mechanic it was the core of the experience.

Performance and What I Would Improve

I believe our app runs all right on the Quest, but definitely can feel the limitations of mobile hardware. If I had more time, I would spend much more of it on user testing. Watch others play, see where they struggle, how they move, when they get stressed - that kind of feedback is gold when you're refining an XR experience.

The Way We Worked as a Team

The other thing that really mattered on the project was the way we worked together. A great deal of our development happened on one big screen, with everyone sitting together and discussing solutions in real time. It was basically informal pair programming, and it made a huge difference.

It is hard to explain with words, you need to see and feel XR. Being able to test directly in the headset, point at things in Unity, and adjust immediately helped us make better decisions and fix problems faster. It gave me a lot more ownership of the project, too, because I wasn't only involved in those parts that I had coded myself, but also in discussions which shaped the whole application.

Personal Growth

The area I have grown the most in is definitely in XR-specific development. I am far more confident with AR Foundation and XR Interaction Toolkit and in building scenes that work well with tracking, interaction, and performance. I also learned to solve problems more systematically instead of trying random fixes. And maybe the biggest change I started to think like an XR developer. Not just "does it work?" but "how does it feel?", "is it intuitive?", "does this make sense for the user's body in 6DOF?", "is this comfortable?". That shift in mindset is something I know I'll take with me into future projects.